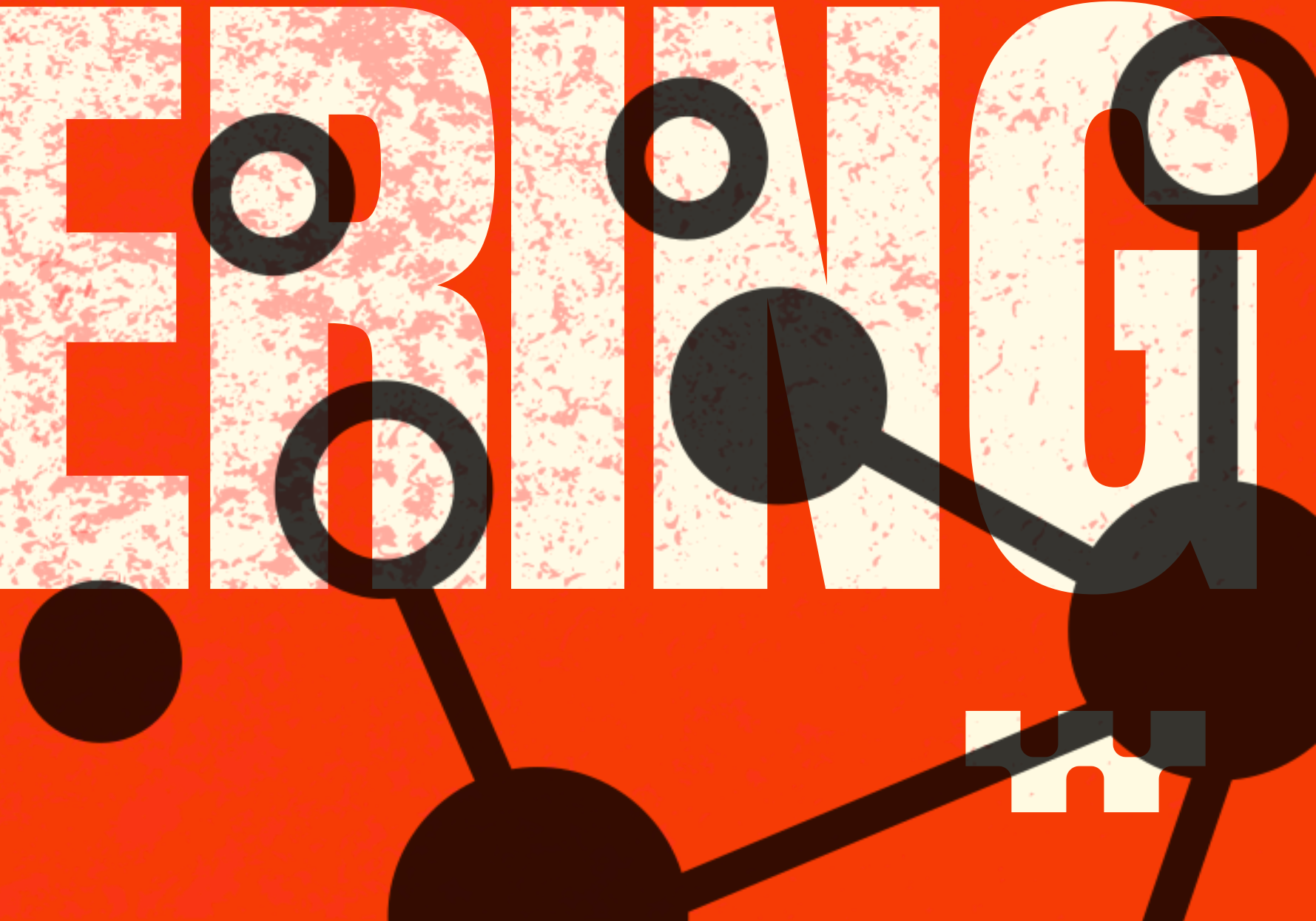


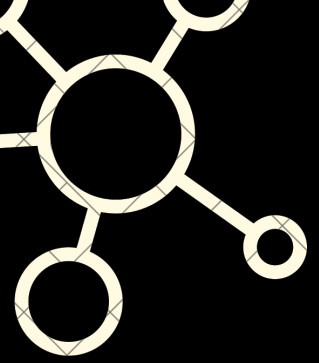
GROUP 15

SELF OPTIMAL

CLUSTERING

OPTIMIZING COLOR IMAGE
SEGMENTATION





INTRODUCTION

CLUSTERING IN HYPERSPACE

NATURE OF CLUSTERS

Collections of homogeneous data points where intra-group similarity is high and inter-group distinction is maximized.

HARD VS FUZZY POSITIONING

Hard clustering assigns an instance to exactly one cluster, whereas fuzzy clustering assigns membership degrees across multiple partitions.

IMPROVED MOUNTAIN CLUSTERING

Our implementation models RGB color features directly as coordinate dimensions in a unit hypercube without relying on underlying physical process assumptions.

K-MEANS & FCM

- Struggle with non-spherical data distributions.
- Highly sensitive to initial partition selection, frequently trapping the execution in local minima.
- Expectation-Maximization (EM) requires massive iteration counts, driving computational complexity to extreme heights.

STANDARD MOUNTAIN CLUSTERING

- Grid-based computation grows exponentially with hyperspace dimensions.
- Destroys the mountain function around selected centers, frequently erasing adjacent points that could serve as valid cluster centers.

IMPROVED MOUNTAIN CLUSTERING

- Resolves center destruction by isolating formed clusters.
- The Fatal Flaw: Relies on heuristically estimated threshold functions, leaving cluster separation mathematically unoptimized.



**LIMITATIONS OF
CURRENT METHODS**

SOC OPTIMIZATION PROTOCOL



STEP 1

Hyperspace Normalization

We correlate δ_m to the Global Silhouette Index (S_t) via Lagrange interpolation:

$$\bar{x}^j = \frac{x^j - (x)_{\min}}{(x)_{\max} - (x)_{\min}}$$

STEP 2

MOUNTAIN POTENTIALS

Potential values P_m^r are computed across Euclidean distances bounded by threshold δ_m :

$$P_m^r = \sum_{j=1}^n \exp \left[- \left(\frac{d^2(\bar{x}^r, \bar{x}^j)}{\delta_m^2} \right) \right]$$

STEP 3

THRESHOLD OPTIMIZATION

We correlate δ_m to the Global Silhouette Index (S_t) via Lagrange interpolation:

$$t = \sum_{m=1}^M S_m \prod_{k=1, k \neq m}^M \frac{(\delta_t - \delta_k)}{(\delta_m - \delta_k)}$$

By finding the roots of this polynomial, the algorithm mathematically pushes the Silhouette Index as close to 1.0 as possible.



MODULAR PYTHON PIPELINE

DATA INGESTION (`main.py`)

- Loads images dynamically via URL or local paths.
- Reshapes 3D RGB matrices into 2D arrays for hyperspace processing.

CORE ALGORITHM (`soc.py`)

- Executes the Improved Mountain Clustering logic.
- Calculates potential values using `numpy` vectorized operations for performance.



MODULAR PYTHON PIPELINE

OPTIMIZER ([factoral.py](#))

- The heart of the algorithm. Iteratively fits the Lagrange polynomial to optimize the threshold factor

VALIDATOR ([slht.py](#))

- Integrates with scikit-learn to compute the Global Silhouette Index (GSI) for performance validation.



- **Initialization:** The threshold scaling factor is set to 1.0 for the first run.
- **Clustering & Scoring:** The SOC algorithm segments the image and calculates the Silhouette Index (St) for the formed clusters.
- **Polynomial Fitting:** The algorithm fits a Lagrange polynomial relating the distance thresholds to the Silhouette scores.
- **Root Extraction:** It extracts the roots of this polynomial to find the exact threshold value that pushes St closest to 1.0.

The 10-Iteration Cap:

To ensure efficiency and fairness, the algorithm is hard-coded to cap at exactly 10 iterations, locking in the threshold that yields the maximum GSI.

ITERATIVE THRESHOLD OPTIMIZATION

PROOF OF CONVERGENCE

GROUP 15

THEORETICAL GROUNDING

- Weierstrass Approximation Theorem: Guarantees that the sequence of interpolating polynomials $S_I(\delta)$ converges uniformly to the true continuous cluster function.
- Chebyshev Alternation Theorem: Proves that selecting nodes at polynomial intersection points yields the exact best-approximation polynomial.

ALGORITHMIC VERIFICATION

- Nelder-Mead Simplex Bounding: Simplex optimization proves the data quotient is strictly bounded within $[0, 0.3333]$, bounding initial threshold δ_1 in $[0, 0.1666]$.
- Dekker's Root-Finding Protocol: Our root extraction simulates Dekker's algorithm—combining bisection, secant, and inverse quadratic interpolation—to ensure robust conditional convergence without divergence.

BENCHMARKING SEGMENTATION QUALITY

GLOBAL SILHOUETTE INDEX (GSI)

- Evaluates if a point is in the right cluster vs. neighboring clusters.
- Goal: Maximize (Closer to 1.0 is better).

SEPARATION INDEX (SI)

- Uses a minimum-distance separation for partition validity.
- Goal: Minimize.

PARTITION INDEX (PI)

- Ratio of the sum of compactness and separation of the clusters.
- Goal: Minimize (Lower value = better partition).

DUNN INDEX (DI)

- Maximises inter-cluster distances while minimising intra-cluster distances.
- Goal: Maximize.

RAW IMAGE



200*200 image

SEGMENTED IMAGE



GSI = 0.429

No. of clusters = 5

QUALITATIVE SEGMENTATION RESULTS

The SOC algorithm successfully demarcates distinct physical regions by grouping homogeneous pixels and sharply separating distinct color channels without physical process assumptions.

QUANTITATIVE RESULTS

Comparing SOC against baseline Improved Mountain Clustering methods (IMC-1 and IMC-2)

METHOD	GSI	PI	SI	DI
IMC-1	0.3557	0.1680	0.3911	0.2657
IMC-2	0.2946	0.2316	0.6072	0.1863
SOC	0.4289	0.1120	0.3419	0.2853

Results on the image with 200*200 pixels and 5 clusters segmented

SOC consistently achieves the highest Global Silhouette Index (GSI) and Dunn Index (DI), proving superior inter-cluster distance.

PERFORMANCE ACROSS COMPLEX TOPOLOGIES

BOTTLE DATASET (M = 4)

- EM GSI: 0.3680
- K-Means GSI: 0.6986
- IMC-2 GSI: 0.6984
- SOC GSI: 0.7056

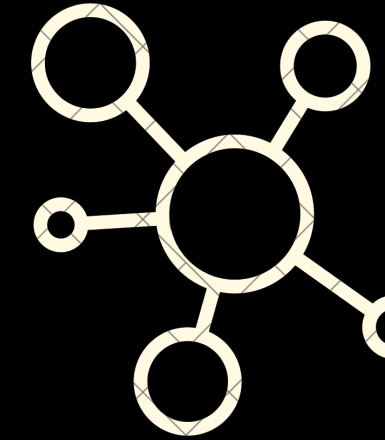
SOC Extracts superior partitions even when ambient lighting degrades heuristic IMC-2

FLOWER DATASET (M = 2)

- EM GSI: 0.8716
- K-Means GSI: 0.8735
- IMC-2 GSI: 0.8731
- SOC GSI: 0.8747

SOC Maintains mathematical dominance across natural organic color boundaries

WHY SOC OUTPERFORMS THE STANDARD !



- **SOLVES FCM'S SENSITIVITY**

Fuzzy C-Means (FCM) is highly sensitive to initial partitions and often gets trapped in local minima. SOC's dynamic thresholding avoids this.

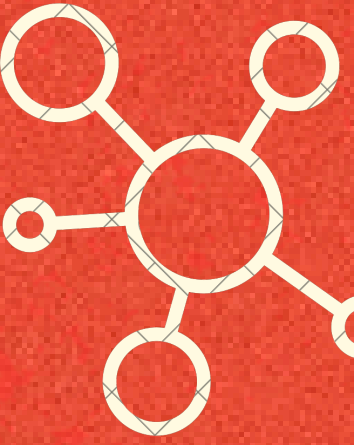
- **ZERO REDUNDANCY**

By isolating and completely eliminating data points once they are assigned to a cluster, SOC finds all relevant clusters without creating redundant groupings.

- **BETTER THAN HEURISTICS**

Attempts to manually tweak the threshold (like "IMC-max" or "IMC-half") fail or create invalid numbers (NaN). Mathematical interpolation is strictly required for peak performance.

CONCLUSION AND FUTURE SCOPE

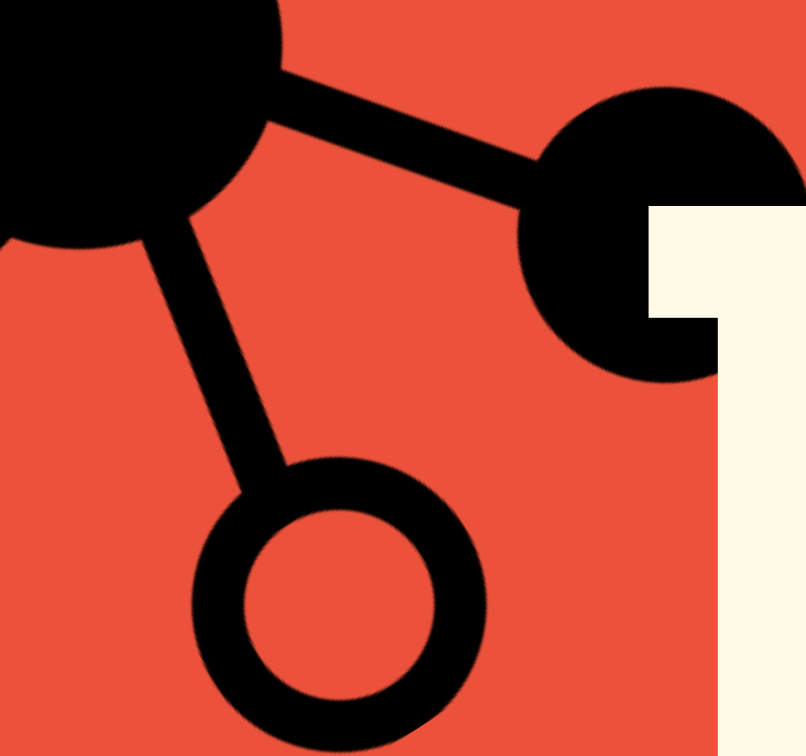


ALGORITHMIC ACHIEVEMENTS

- Eliminates manual parameter guessing by introducing self-converging threshold optimization.
- Overcomes legacy FCM local-minima trapping and extreme EM time complexity.

APPLICATION DOMAINS

- Medical Diagnostic Imaging: Demarcating distinct anatomical structures and tumor boundaries in multi-spectral radiography with absolute precision.
- Bioinformatics & Genomics: Analyzing high-density gene expression profiles and microarray datasets where isolating non-overlapping clusters is critical.



THANK YOU

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SCAN FOR CODE



GROUP 15

